Subject: Mathematics | **WEEKLY CHAPTER TEST – 04 (LINEAR EQUATIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 9(iv).
2. **Section A** comprises 3 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 1 Assertion-Reason question of 1 mark.
4. **Section C** comprises 2 Very Short Answer (VSAQ) questions of 2 marks each.
5. **Section D** comprises 1 Short Answer (3M) and 1 Long Answer / HOTS (5M) question.
6. **Section E** comprises 1 Applied Case Study with 4 sub-parts of 1 mark each.

SECTION A: Multiple Choice Questions (3 × 1 = 3 Marks)

Q1. The sum of the digits of a two-digit number is 9. If 27 is added to it, the digits get reversed. The number is: [1]

(A) 25 (B) 72 (C) 63 (D) 36

Q2. Aruna has only Rs. 1 and Rs. 2 coins with her. If the total number of coins is 50 and the amount of money is Rs. 75, then the number of Rs. 1 and Rs. 2 coins are, respectively: [1]

(A) 35 and 15 (B) 35 and 20 (C) 15 and 35 (D) 25 and 25

Q3. If $x = a, y = b$ is the solution of the equations $x - y = 2$ and $x + y = 4$, then the values of a and b are: [1]

(A) 3 and 1 (B) 3 and 5 (C) 5 and 3 (D) -1 and -3

SECTION B: Assertion-Reason Question (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The area of the rectangle formed by the lines representing $x = 8, y = 6$ with the coordinate axes is 48 sq. units.
Reason (R): The lines $x = 8$ and $y = 6$ intersect at $(8, 6)$, forming a right rectangle of length 8 units and breadth 6 units with the coordinate axes $x = 0$ and $y = 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: Very Short Answer Questions (2 × 2 = 4 Marks)

Q5. Given $217x + 131y = 913$ and $131x + 217y = 827$, find the values of $(x + y)$ and $(x - y)$, and hence find the solution (x, y). [2]

Q6. Determine whether the lines representing $2x - 3y = 8$ and $4x - 6y = 9$ are intersecting, coincident, or parallel. [2]

SECTION D: Short & Long Answer Questions (1 × 3 + 1 × 5 = 8 Marks)

Q7. The taxi charges in a city consist of a fixed charge together with the charge for the distance covered. For a distance of 10 km, the charge paid is Rs. 105 and for a journey of 15 km, the charge paid is Rs. 155. What are the fixed charges and the charge per km? How much does a person have to pay for travelling a distance of 25 km? [3]

Q8. A person invested some amount at 12% simple interest per annum and another amount at 10% simple interest per annum. He received a total yearly interest of Rs. 130. But if he had interchanged the amounts invested, he would have received Rs. 4 more as yearly interest. How much amount did he invest at each rate? [5]

SECTION E: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q9. Case Study: Theme Park Ticketing Gate Analytics

On a public holiday, a popular amusement theme park offered special admission passes. The entry ticket for a child is priced at Rs. x and for an adult is priced at Rs. y.

Exactly 100 children and 200 adults entered the park that day. The cashier collected a total sum of Rs. 60000. It is known that an adult ticket costs Rs. 150 more than a child ticket, i.e., $y - x = 150$.

Based on the above information, answer the following questions:

(i) Formulate the pair of linear equations in x and y representing the total collection and the price difference. [1]

(ii) Find the individual ticket price for a child (x) and for an adult (y). [1]

(iii) Verify the total number of visitors who entered the park. [1]

(iv) If the park offers a 20% festive discount on adult tickets next Sunday, find the revenue collected from 100 children and 200 adults. [1]

OR

Find the total revenue if 150 children and 150 adults visit at standard rates. [1]

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